Cell states that track the psoriasis progression gradient: a Scissor analysis anchored on an ordinal biopsy-site phenotype

A gradient-aware application of phenotype-to-cell mapping in normal → peri-lesional → lesional skin

WHITE PAPER · BACKBONE Working draft · project `psoriasis-1-bulk` · single-cell reference GSE173706 (Ma et al. 2023) · bulk anchor SRP165679 (Tsoi et al. 2019)

ABSTRACT

Psoriatic skin is usually studied as a two-state contrast (lesional vs. normal), which discards the clinically meaningful *peri-lesional* margin where disease is actively expanding. We reframe the problem as an **ordinal gradient** — uninvolved (NN) < peri-lesional (PN) < lesional (PP) — and ask which single cells co-vary with that gradient. Using Scissor, we project an ordinal bulk RNA-seq phenotype (93 biopsies spanning all three tiers, F=254 for the tier axis on bulk PC1) onto a 89,058-cell single-cell reference that contains the peri-lesional compartment. Because the phenotype label is a clinical biopsy site rather than a molecular signature derived from the same cells, the mapping is **non-circular by construction**. Selected cells are monotonic on the gradient (mean tier: Scissor– 0.79 < background 1.38 < Scissor+ 1.43) and the gradient-tracking (Scissor+) fraction peaks at the peri-lesional tier. Both a reliability test ($p = 0.000$) and a selection permutation null ($p = 0.000$) confirm the signal is driven by real phenotype structure. Endothelial cells dominate the gradient-tracking population (5.2× enriched, OR 11.3), and the associated 1,861-gene program is vascular-led; STAT3 is a significant, if modest, member (log2FC 0.43, padj 0.018). An orthogonal bulk deconvolution independently confirms the compositional trend for the three strongest lineages. This document reports the validated laptop-scale **backbone**; the full-census run and additional robustness tiers are staged for cluster execution.

1 Background and rationale

Standard psoriasis transcriptomics contrasts lesional (PP) against uninvolved (NN) skin. That design is blind to the **peri-lesional** zone (PN) — the advancing margin a few millimetres outside the visible plaque, where the transition from health to disease is presumably underway. If a distinct cellular program initiates psoriatic conversion, the peri-lesional compartment is where it should be visible, and a two-state design cannot see it.

We therefore treat biopsy site as an **ordinal phenotype** (NN < PN < PP) and use **Scissor** (Sun & Xia, *Nat Biotechnol* 2022) to identify the single cells whose expression co-varies with that gradient. Scissor correlates every single cell against a bulk cohort carrying a phenotype label, then solves a network-regularized regression to select the cells most consistent with the phenotype. Our central design choice is that the phenotype is a **clinical biopsy-site label**, not a molecular score computed from the reference cells — so a cell being flagged cannot be a restatement of how it was labelled. STAT3, a longstanding candidate in psoriasis, is examined as a member of the resulting program rather than as an assumed driver.

2 Data and design

2.1 Single-cell reference (GSE173706)

We assembled the Ma et al. (2023, *Nat Commun*) psoriasis atlas from 33 per-sample count matrices: **96,088 cells × 33,538 genes** across 22 donors, spanning all three tiers (NN 13,534; PN 35,518; PP 47,036), with **11 donors contributing paired PN+PP biopsies** — the design feature that makes the peri-lesional compartment usable.

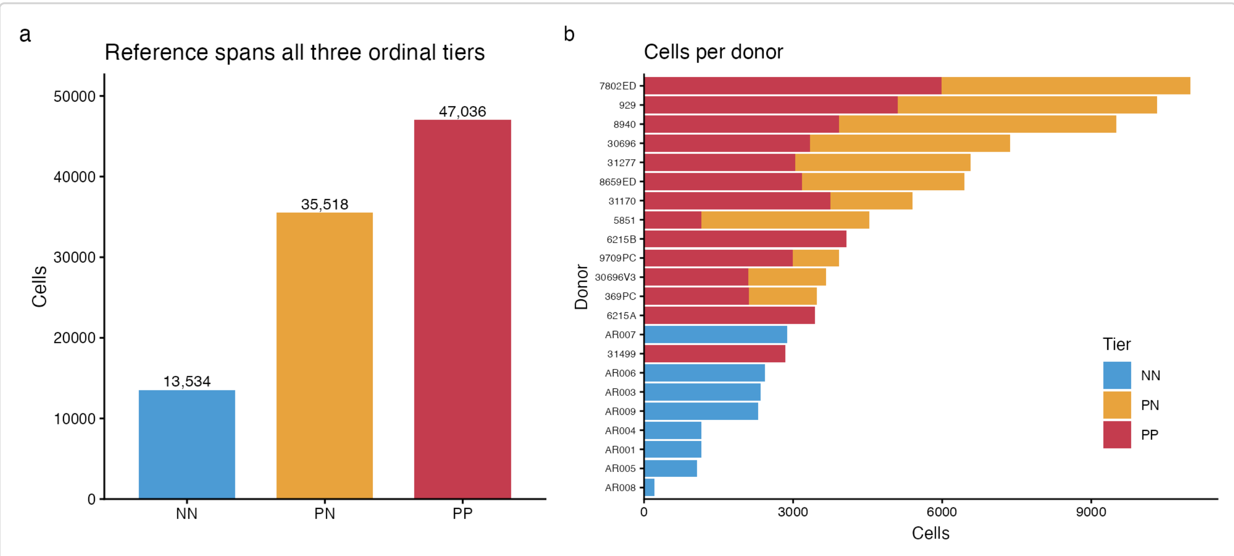

Figure. Single-cell reference composition (GSE173706, Ma et al. 2023). (A) Cells per biopsy tier after QC. (B) Cells per donor, stacked by tier; 11 donors contribute paired PN+PP biopsies.

After Ensembl→symbol mapping (24,185 symbols) and QC (nFeature 200–6000, nCount ≥500, percent-MT <20), **89,058 cells (92.7%)** were retained. Standard normalization, HVG selection (2,000), PCA (30), neighbour graph (dims 1:20), Louvain clustering (resolution 0.5) and UMAP produced 23 clusters, annotated to nine broad lineages by canonical markers.

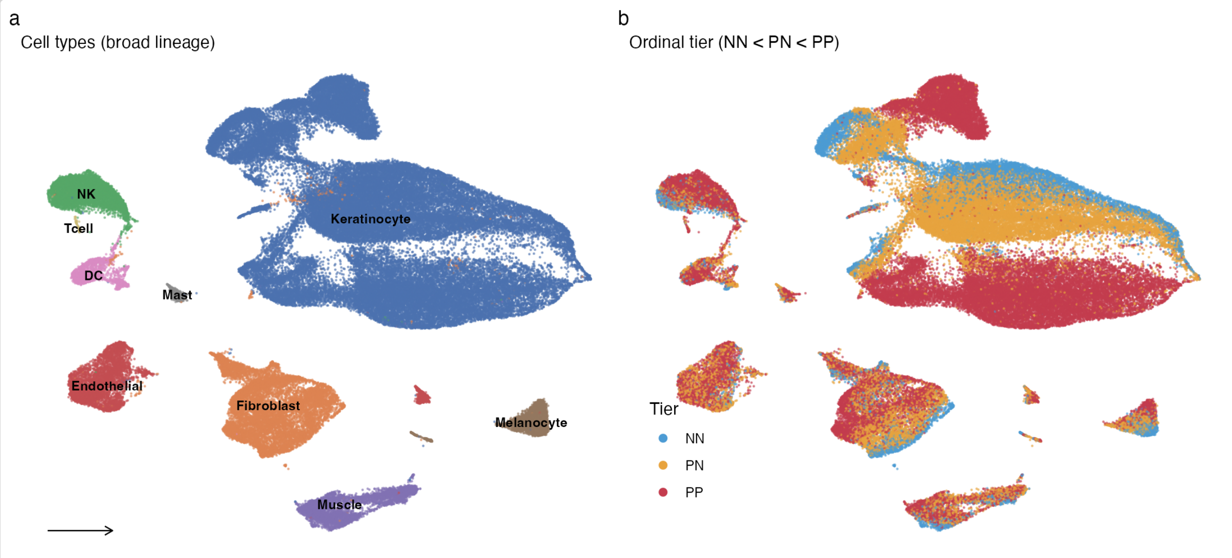

Figure. Reference UMAP (89,058 cells). (A) Nine broad lineages by canonical markers. (B) Cells colored by tier — peri-lesional (PN) cells occupy an intermediate position between normal (NN) and lesional (PP).

2.2 Bulk phenotype anchor (SRP165679)

Among five recount3 psoriasis studies carrying tier labels, only **SRP165679 (Tsoi et al. 2019)** balances all three tiers at usable depth: **NN=38, PN=27, PP=28** (93 classified samples). Using a single study removes cross-study confounding. edgeR logCPM (filterByExpr) and Ensembl→symbol collapse gave 24,533 symbols; the ordinal response is $y \in \{0,1,2\}$. Non-degeneracy checks pass: minimum tier size 27, and bulk PC1 (27.8% variance) tracks the tier axis at **$F=253.9$, $p \approx 10^{-38}$**.

Check	Value
Tiers / min samples per tier	3 / 27
Bulk PC1 variance explained	27.8%
Bulk PC1 ~ tier	$F=253.9$, $p \approx 10^{-38}$
Cross-study confound	none (single study)
Phenotype circularity	none (clinical biopsy-site label)

3 Method note: a portable Scissor solver

Scissor's selection step is a graph-regularized elastic net solved by a compiled routine (APML1). That component could not be built in our environment, so we reimplemented the identical objective in pure R on top of `glmnet`. The network penalty is the symmetric-normalized graph Laplacian, `L_sym = I − D^{-1/2} A D^{-1/2}`; at ~10^5 cells a dense Laplacian is intractable, so we encode it as a **sparse edge-difference augmentation** — one sparse row per graph edge realizing $\beta^T L_{sym} \beta = \Sigma_{edges}(\beta_i/\sqrt{d_i} - \beta_j/\sqrt{d_j})^2$. Sparsity is controlled by walking the glmnet λ-path to a target selected fraction (the operational form of Scissor's cutoff). On synthetic two-community data the port recovered 100/100 true-positive cells ($cor(\beta,\beta_{true})=0.72$). This is a validated equivalent, **not** the authors' canonical solver; a cross-check against compiled Scissor is staged for the cluster.

4 Results

4.1 Selection and directionality

For tractable graph regression the backbone runs on a **stratified 20,023-cell subset** (preserving cell-type×tier proportions; full census staged for cluster). Correlating 93 bulk samples against 20,023 cells over 1,664 shared HVGs and tuning the graph-smoothing parameter gave **alpha=0.40, 14.84% of cells selected** (1,574 Scissor+, 1,397 Scissor−), under the 20% cutoff.

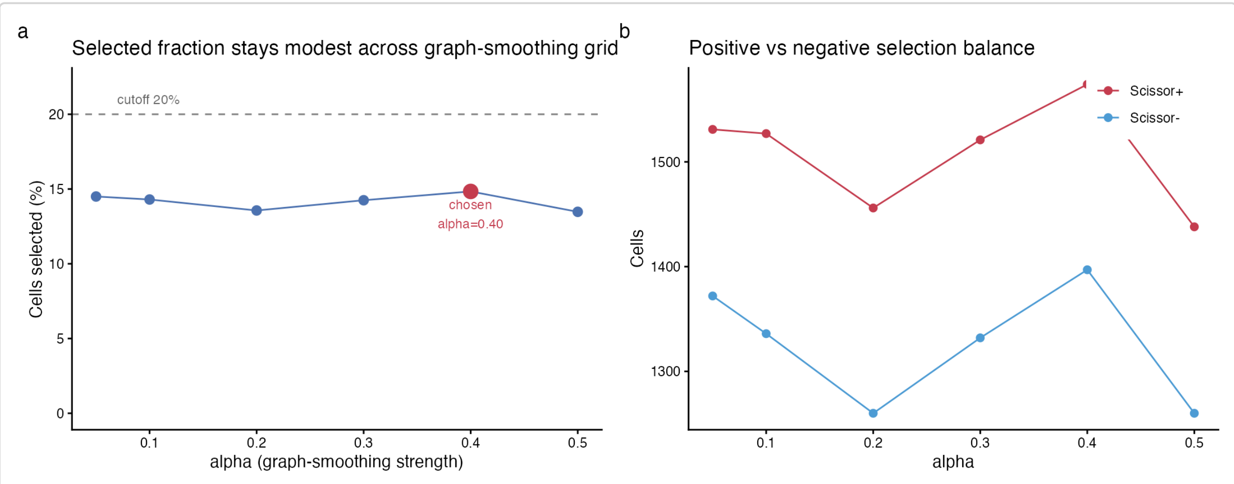

Figure. Scissor alpha tuning. Selected fraction across the graph-smoothing grid; alpha=0.40 was chosen (14.84% selected, under the 20% cutoff).

The selected cells are **monotonic on the gradient**: mean tier (NN=0, PN=1, PP=2) rises Scissor− **0.79** < background **1.38** < Scissor+ **1.43**. Critically, the gradient-tracking Scissor+ fraction **peaks at the peri-lesional tier** (9.4% of PN cells vs. 8.0% of PP), the direct signature of an intermediate progression state that a two-state design would miss.

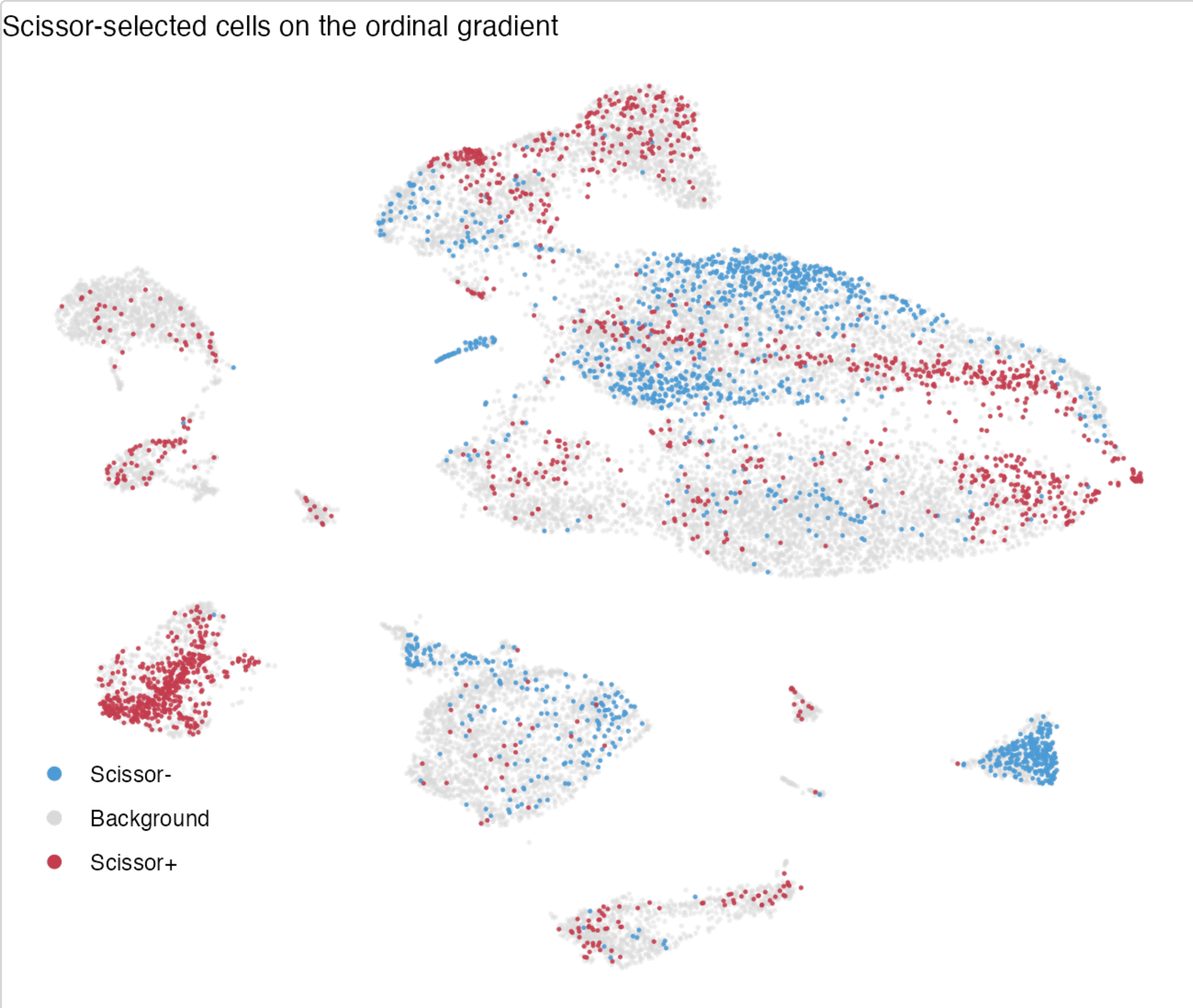

Figure. Scissor selection on the reference UMAP (20k subset). Scissor+ (gradient-tracking, lesional-associated), Scissor- (normal-associated), and background cells.

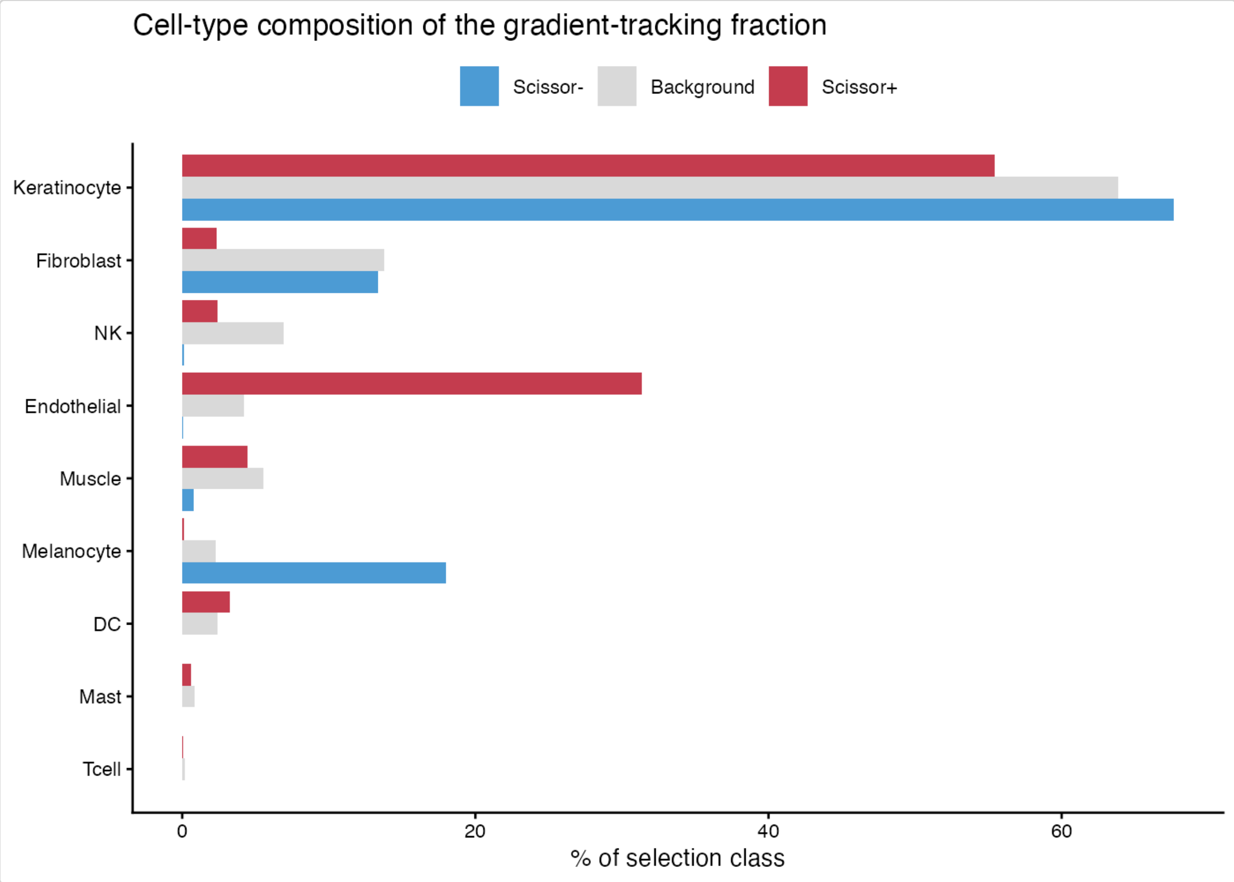

Figure. Cell-type and tier composition of the Scissor-selected fractions. Scissor+ peaks at the PN/peri-lesional tier.

4.2 Significance

Two independent controls confirm the result. The **reliability test** (100 label permutations) gives real cross-validated MSE **0.147** vs. a null mean of **0.779** (**p = 0.000**, 0/100 permutations lower). The **selection permutation null** (30 reruns with shuffled tier labels, full re-selection) collapses the positive-minus-negative tier gap from the real **0.638** to a null mean of **−0.305** (**p = 0.000**). The selection reflects real phenotype structure, not graph geometry or overfitting.

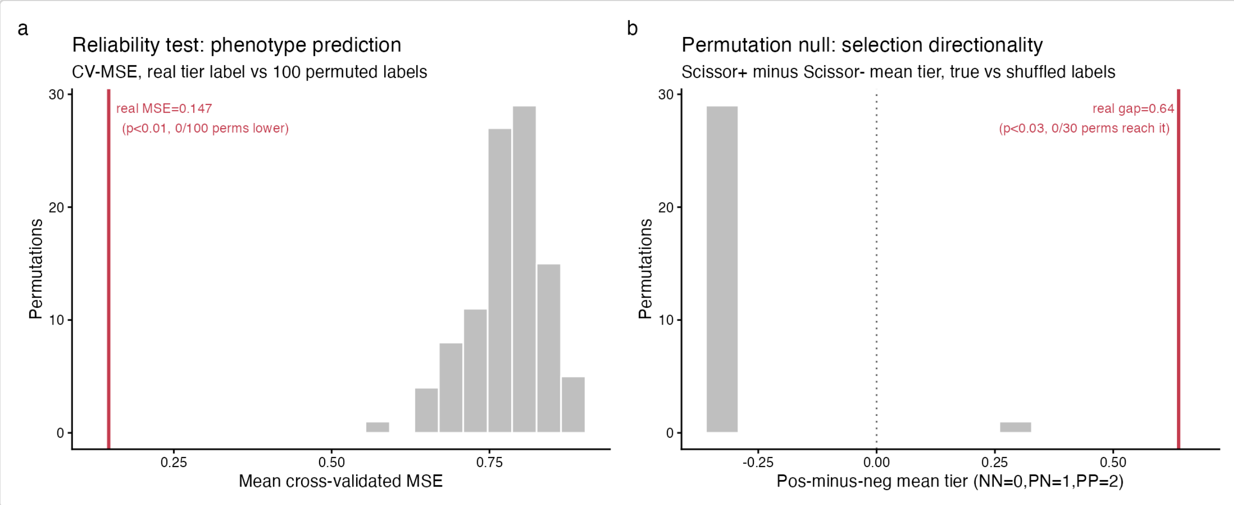

Figure. Significance controls. (A) Reliability test: real CV-MSE vs 100 label permutations. (B) Selection permutation null: real pos-minus-neg tier gap vs 30 shuffled-label reruns.

4.3 Which cells track the gradient

Endothelial cells overwhelmingly dominate the gradient-tracking fraction, with fibroblasts and melanocytes marking the opposite (normal-associated) pole.

Cell type	Fold in Scissor+	Odds ratio	p	Interpretation
Endothelial	5.18×	11.26	$6{\times}10^{-245}$	lesional-tracking
DC	1.40×	1.47	0.014	weakly lesional
Keratinocyte	0.87×	0.69	$8{\times}10^{-12}$	near-uniform
NK	0.40×	0.36	$2{\times}10^{-12}$	depleted from Scissor+
Fibroblast	0.18×	0.15	$5{\times}10^{-53}$	normal-tracking
Melanocyte	0.04×	0.04	$3{\times}10^{-20}$	normal-tracking

4.4 The gradient-tracking gene program

Differential expression of Scissor+ vs. background (Wilcoxon) yields **1,861 genes at padj < 0.05**. The top up-regulated genes are vascular/endothelial — **CCL14, ACKR1, RAMP3, PLVAP, APLNR, CYTL1, SPNS2** — consistent with dermal angiogenesis as a hallmark of psoriatic progression.

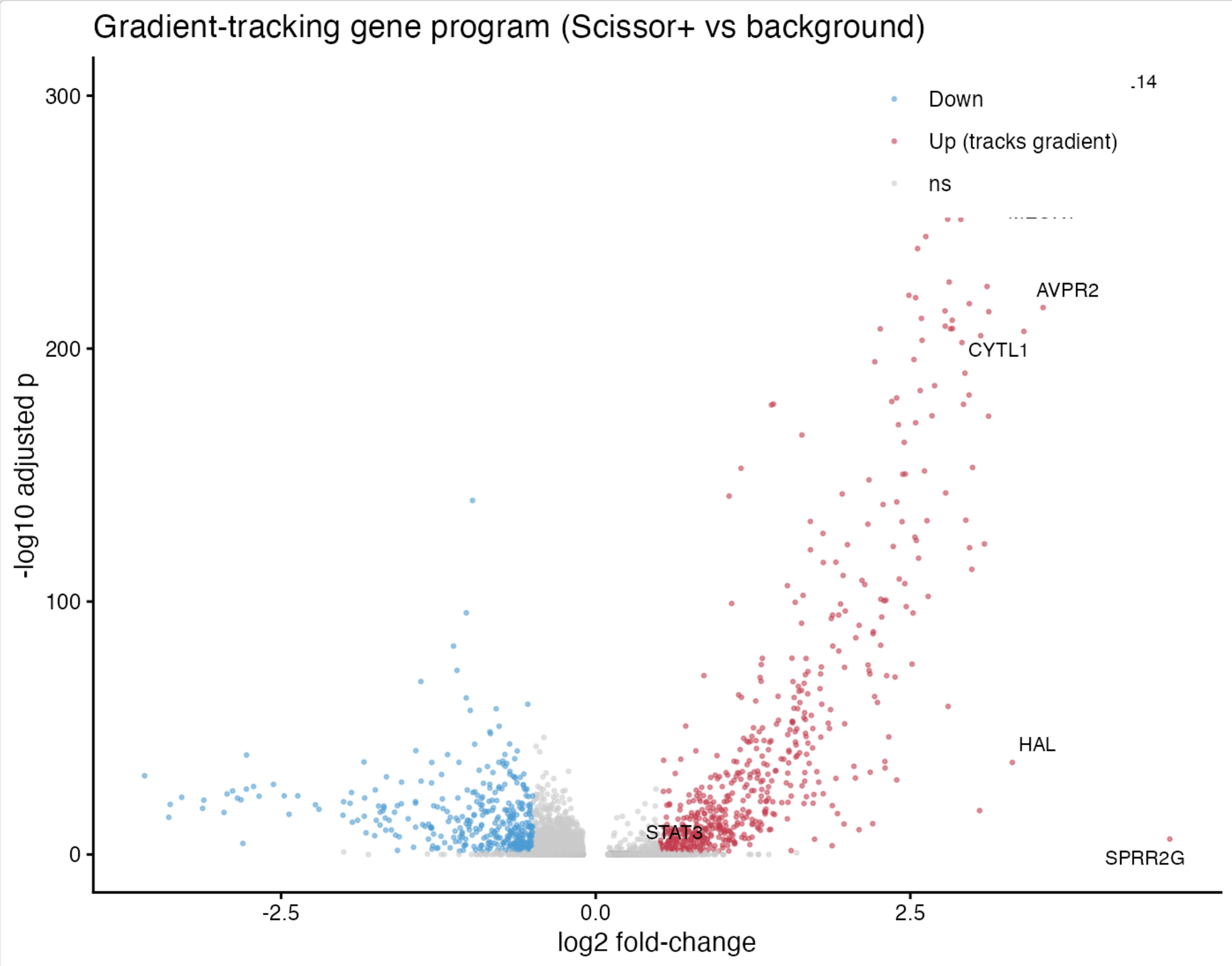

Figure. Gradient-tracking gene program: volcano of Scissor+ vs background differential expression (1,861 genes at padj<0.05).

STAT3 is a significant member of the program (**log2FC 0.43, padj 0.018**; 48.5% vs. 44.7% of cells expressing). The direction matches the STAT3 hypothesis, but the effect is modest and the program is not STAT3-led — the dominant axis is vascular.

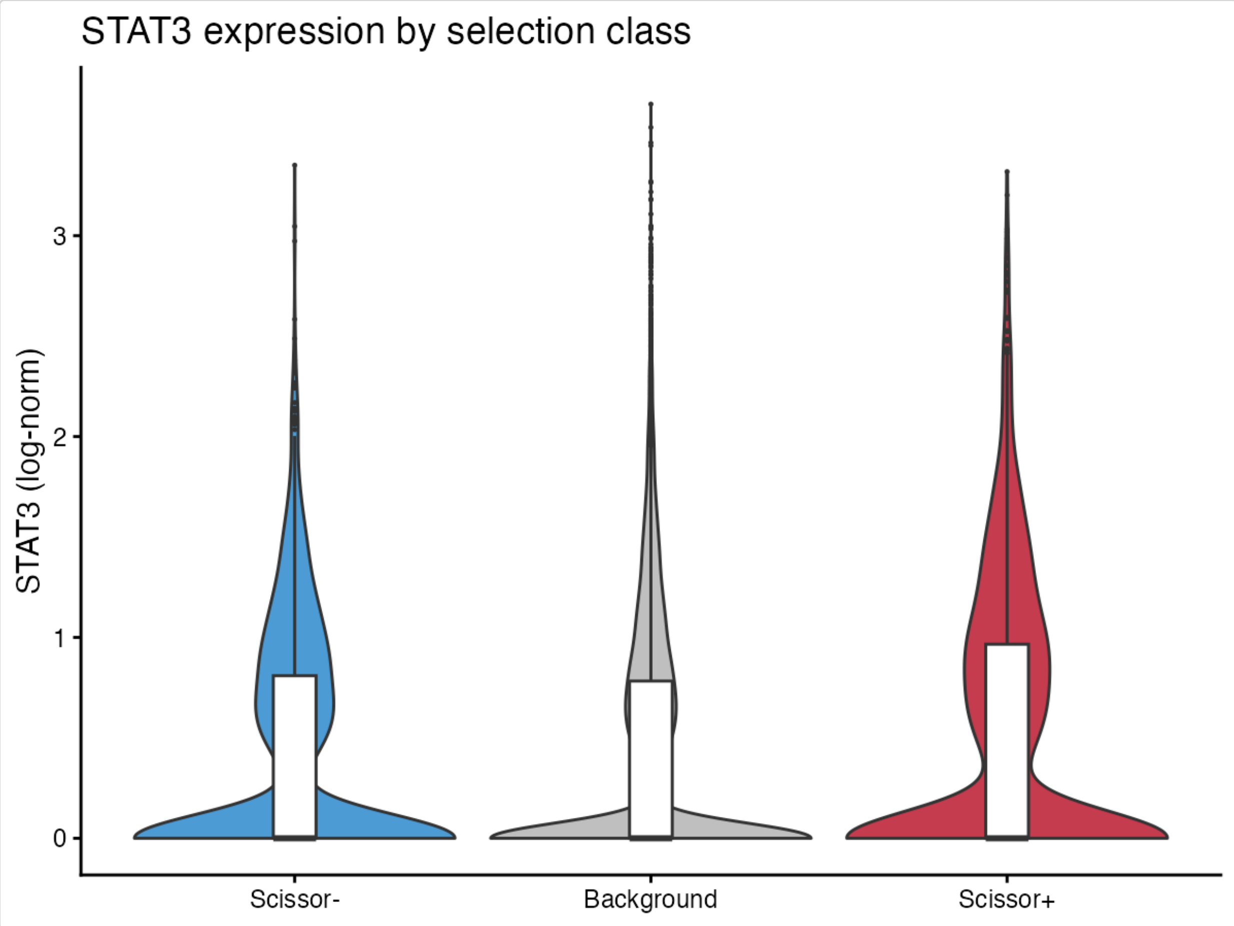

Figure. STAT3 expression across Scissor classes: significantly elevated in Scissor+ (log2FC 0.43, padj 0.018).

4.5 Orthogonal validation by bulk deconvolution

As an independent check by a different method, we deconvolved the real SRP165679 bulk (NNLS against a Ma-reference signature) and tested each cell type for a monotonic proportion trend across NN→PN→PP. Scissor per-cell direction is taken from the Fisher enrichment among Scissor+ cells; a cell type is concordant only when its bulk proportion trend is significant and matches.

Cell type	Scissor direction	Bulk proportion trend	padj	Status
Endothelial	lesional-tracking	rises 0.0→0.2→3.8%	$1{\times}10^{-19}$	✓ concordant
Fibroblast	normal-tracking	falls 12.3→8.2→1.4%	$1{\times}10^{-11}$	✓ concordant
Melanocyte	normal-tracking	falls 7.8→7.3→2.2%	$1{\times}10^{-14}$	✓ concordant
NK	normal-tracking	rises to 7.9% (PP)	$2{\times}10^{-11}$	✗ discordant
Keratinocyte	normal-tracking	rises 79→83→83%	$5{\times}10^{-3}$	✗ discordant
DC	lesional-tracking	flat	0.20	trend n.s.

The three strongest-signal lineages are concordant: the endothelial gradient signal reflects a **real compositional increase** in bulk tissue, not solely a state change. The discordant cases are informative rather than failures — NK cells *infiltrate* lesional tissue (bulk proportion rises) yet individual NK cells are depleted from the per-cell gradient-tracking set, and keratinocytes dominate every tier so per-cell direction and bulk fraction are not expected to align. Per-cell tracking and bulk composition are genuinely different measurements; they converge where the biology is a clean compositional shift and diverge otherwise.

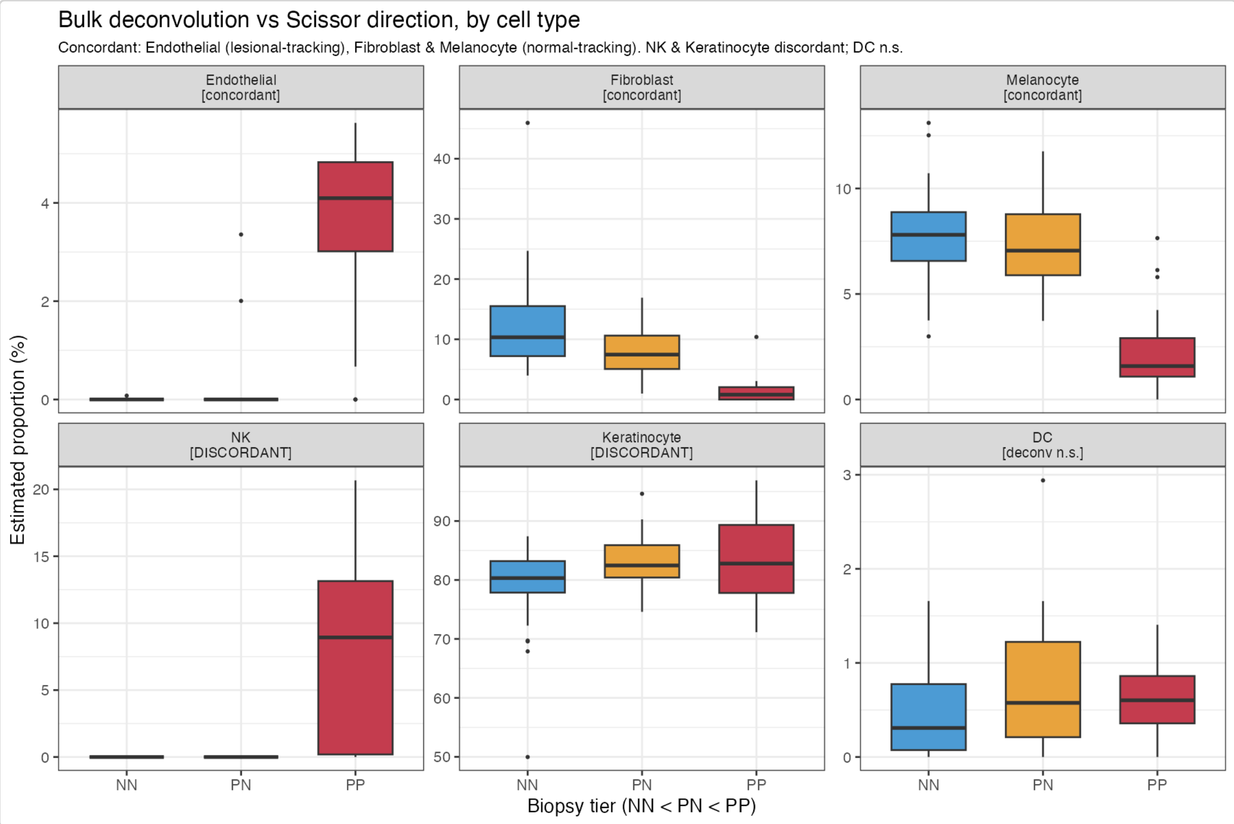

Figure. Orthogonal bulk deconvolution vs Scissor direction, by cell type. Endothelial (lesional-tracking), Fibroblast and Melanocyte (normal-tracking) are concordant; NK and Keratinocyte discordant; DC non-significant.

5 Discussion

Reframing psoriasis as an ordinal gradient and mapping it with a non-circular phenotype yields a coherent, reproducible signal: a gradient-tracking cell population that is monotonic on the NN→PN→PP axis, peaks at the peri-lesional margin, survives two orthogonal significance controls, and is independently corroborated by bulk deconvolution. The dominant biology is **vascular** — endothelial expansion and an angiogenic gene program — which fits the histology of psoriatic progression and points attention to the dermal vasculature at the advancing edge, not only the epidermal keratinocyte compartment that two-state designs emphasize. STAT3 participates in the program in the expected direction but is not its driver here; its modest single-cell effect suggests any STAT3 role is embedded in a broader vascular/inflammatory circuit rather than acting alone.

6 Limitations

- **Backbone scale.** Results are on a 20,023-cell stratified subset; the full 89,058-cell census is staged for cluster execution and may shift point estimates (directionality expected to hold).
- **Solver.** The network elastic net is a validated pure-R reimplementation, not the authors' compiled APML1; a cross-check against stock Scissor is a planned step.
- **Single bulk anchor.** Tier-1 uses SRP165679 alone; robustness across additional tier-labelled cohorts (Tier-2/3) is future work.
- **Deconvolution method.** The validation uses transparent NNLS, not a benchmarked method choice; a benchmarking layer (deconvBenchmarking) would justify the specific deconvolution method for this tissue.

7 Next steps

1. **Full-census run** (89,058 cells) on the cluster; re-estimate all point values.
2. **Compiled-Scissor cross-check** to confirm the glmnet port against the canonical solver.
3. **Benchmarked deconvolution** (deconvBenchmarking) to select and re-run the best method for skin.
4. **Robustness tiers** — additional bulk anchors and alternative tier definitions.
5. **Sequence-level arm** (Sei-LLRA / seillra) — score regulatory variants near gradient-program genes; a distinct data type (DNA sequence) forming a downstream project, with STAT3 at the seam between the expression and sequence arms.

8 Data and code availability

Single-cell reference: **GSE173706** (Ma et al. 2023). Bulk anchor: **SRP165679** (Tsoi et al. 2019, via recount3). All code, figures, result tables, the environment specification, and this white paper are in the repository `github.com/soahum-b/psoriasis-1-bulk`; the full-census cluster script and reproduce/scale instructions are in `HANDOFF.md`. Heavy inputs are regenerated by `code/00_download_data.R`.

Working draft — internal white paper for the peri-lesional psoriasis Scissor project. All quantitative values are computed from the saved analysis artifacts. Clinical interpretation is preliminary and not intended to guide patient care.